

ANALYSIS OF LINE COMMUTATED AND VOLTAGE SOURCE CONVERTERS

2

IMPORTANT QUESTIONS

PART-A

Q.1 What is firing angle?

Ans. Firing Angle : The angle at which thyristor is triggered. It is defined as the angle between the zero crossing of the input voltage and the instant that thyristor is fired.

Q.2 What is meant by pulse number of a converter?

Ans. Pulse Number of Converter : It is defined as the ratio of the base frequency of the DC voltage ripple to the fundamental frequency of the AC voltage is called pulse number.

Q.3 Define overlap and overlap angle.

Ans. Overlap and Overlap Angle : Overlap is the phenomenon due to the effect of source inductance on the a.c. side. The current commutation is delayed due to the source inductance which is normally the leakage reactance of a transformer. The waveforms with commutation period, denoted by μ during which both the outgoing diode and incoming diode are conducting. This period is also known as "overlap" period.

In other words, the commutation period, when outgoing and incoming thyristors are conducting, is also known as the overlap period. The angular period both devices share conduction is known as the commutation angle or overlap angle.

Q.4 Write the assumptions for analysis of 6-pulse converter?

Ans. Assumptions for Analysis of 6-pulse Converter :

To consider the theoretical analysis of a conventional 6-pulse bridge, the following assumptions are made:

1. DC current is constant (i.e. the smoothing reactor is infinite).
2. Valves are ideal switches.
3. AC system is infinitely strong (i.e. the 3 phase emfs are balanced and perfectly sinusoidal).

Q.5 Why series and parallel operation of thyristor in HVDC transmission?

Ans. Reasons for Series and Parallel Operation of Thyristor in HVDC Transmission :

- (i) For higher voltage rating use series operation of thyristor.
- (ii) For higher current rating use parallel operation of thyristor.

Q.6 Mention the various modes of characteristics of 12-pulse converter.

Ans. Various Modes of Characteristics of 12-pulse Converter :

Mode 1 : 4 and 5 valve conduction $0 < u < 30$ deg.

Mode 2 : 5 and 6 valve conduction $30 \text{ deg.} < u < 60 \text{ deg.}$

Mode 3 : 6 valve conduction $0 < \alpha < 30 \text{ deg.}, u = 60 \text{ deg.}$

Mode 4 : 6 and 7 valve conduction $60 \text{ deg.} < u < 90 \text{ deg.}$

Mode 5 : 7 and 8 valve conduction $90 \text{ deg.} < u < 120 \text{ deg.}$

Q.7 What is a 12-pulse converter?

Ans. 12-Pulse Converter : A 12 pulse converter is obtained by the series connection of two bridges. The 30° phase displacement between the two sets of source voltages is achieved by the transformer connections, Y/Y for feeding one bridge and Y/ Δ for feeding the second bridge.

Q.8 Draw the output voltage vs firing angle characteristic of a three phase rectifier.

Ans. Output Voltage vs Firing Angle :

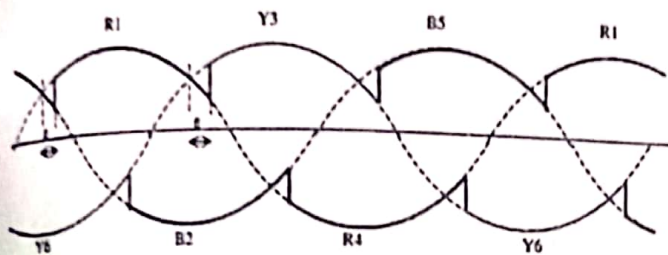


Fig. : Thyristor voltage waveforms

Q.9 Why is 3-phase bridge connection used invariably in converter and inverter circuits in HVDC systems?

Ans. Three phase circuits are preferable when large power is involved. The controlled rectifier can provide controllable output dc voltage in a single unit. Control over the output dc voltage is obtained by controlling the conduction interval of each thyristor. Since thyristors can block voltage in both directions. It is possible to reverse the polarity of the output dc voltage and hence feed power back to the ac supply from the dc side.

PART-B

Q.10 What is a Graetz bridge?

Ans. Graetz Bridge : The basic configuration of a three phase converter (both LCC and VSC) is a bridge converter (also called Graetz bridge) which can be fed from transformer windings connected in star or delta. The converter transformer feeding a Graetz bridge serves the objectives of providing :

- Galvanic separation between AC and DC sides.
- Voltage transformation between AC and DC networks.
- Adjustment of the applied AC voltage by On Load Tap Changer (OLTC).

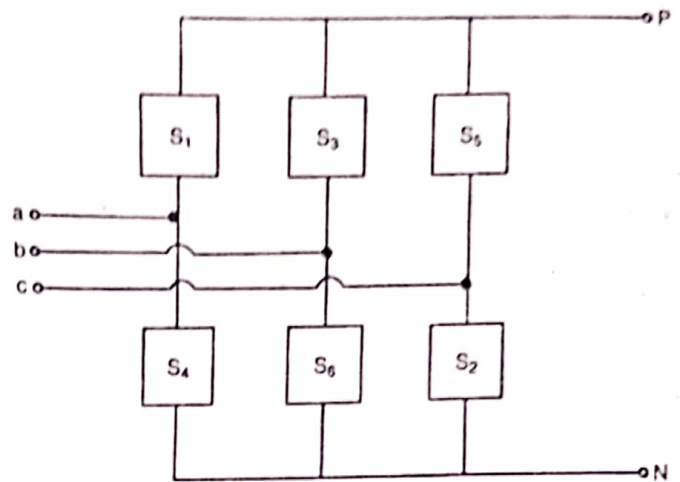


Fig. : A Graetz bridge

Note that an autotransformer can meet the last two objectives but cannot provide galvanic separation. The Graetz bridge circuit is shown in fig.

There are 6 switches numbered in the order in which they are turned on. In a LCC, the switches are made of thyristor valves, whereas in a VSC, they are made up of IGBT valves.

Q.11 Explain in detail about the choice of converter configuration with valve rating and transformer rating.

Ans. Choice of Converter Configuration for any Pulse Number : Graetz bridge is a six pulse converter for which the lowest DC voltage harmonic is sixth. Correspondingly, lowest AC current harmonics are 5th and 7th. To reduce the harmonic content in the AC current and DC voltage, it is desirable to use higher pulse numbers. In general, it can be stated that the characteristic harmonics (under ideal conditions) are of the order

$$h_{dc} = np, h_{ac} = np \pm 1 \quad \dots(i)$$

Where 'n' is an integer, 'p' is the pulse number.

There are several configurations for a converter of a specified pulse number. For $p = 6$, we have in addition to the Graetz bridge, six-phase diametral connection, cascade of three single phase full wave converters, cascade of two three phase converter, parallel connection with interphase transformer etc.

It is convenient to consider a 'p' pulse converter made up of series and parallel connections of a basic valve (commutation) group of 'q' valves or switches as shown in fig. (1). Here, the switch is a thyristor valve whose firing can be delayed (from the instant when the valve voltage becomes positive). The voltage sources are actually obtained from the converter transformer windings. Neglecting overlap, only one valve conducts in a commutation group of 'q' valves.

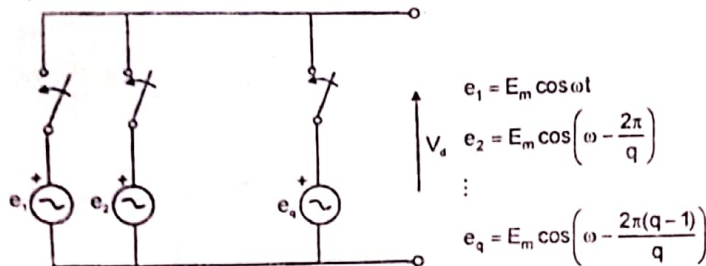


Fig. 1 : A valve group with 'q' valves

If the converter is made up of a matrix of 's' valve groups in series and 'r' valve groups in parallel, then,

$$p = qrs \quad \dots(ii)$$

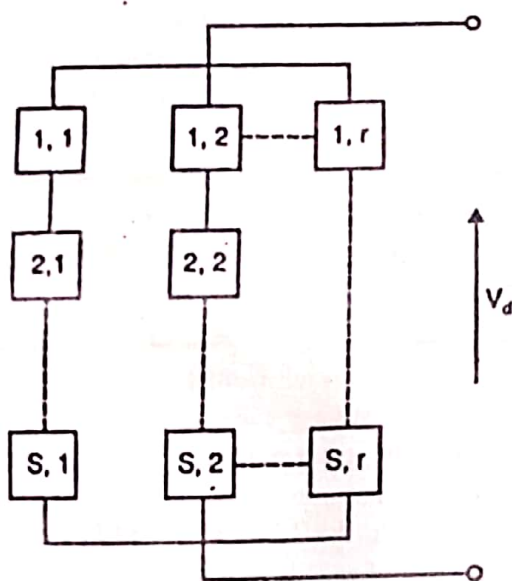


Fig. 2 : Converter made up of series and parallel connection of commutation groups

See fig. (2) for the converter configuration. In general, there are 'p' transformer windings. It will be shown later that sometimes windings can be combined (in particular for $q = 3, r = 1, s = 2$).

Valve Rating : The valve voltage rating is specified in terms of peak inverse voltage (PIV) it has to withstand. The ratio of PIV to the average dc voltage is an index of the valve utilization. The average maximum dc voltage across the converter is given by

$$V_{do} = s \frac{q}{2\pi} \int_{-\pi/q}^{\pi/q} E_m \cos \omega t d\omega t$$

$$= \frac{sq}{\pi} E_m \sin \frac{\pi}{q} \quad \dots(iii)$$

The peak inverse voltage (PIV) across a valve can be obtained as follows:

If 'q' is even, then the maximum inverse voltage occurs when the valve with a phase displacement of π radian (180°) is conducting and this is given by

$$PIV = 2E_m$$

If 'q' is odd, maximum inverse voltage occurs when the valve with a phase shift of $\pi \pm \pi/q$ is conducting. In this case,

$$PIV = 2E_m \cos \frac{\pi}{2q} \quad \dots(iv)$$

The valve utilization factor is given by

$$\frac{PIV}{V_{do}} = \frac{2\pi}{sq \sin \frac{\pi}{q}} \text{ for } q \text{ even}$$

$$= \frac{\pi}{sq \sin \frac{\pi}{2q}} \text{ for } q \text{ odd}$$

Table shows the valve utilization factor for different six-pulse converter configurations. The best valve utilization is obtained for configurations 1 and 3.

Table : Valve Utilization Factor

S. No.	q	r	s	$\frac{PIV}{V_{do}}$
1.	2	1	3	1.047
2.	2	3	1	3.142
3.	3	1	2	1.047
4.	3	2	1	2.094
5.	6	1	1	2.094

Transformer Rating : The current rating of a valve (as well as transformer winding supplying it) is given by

$$I_v = \frac{I_d}{r\sqrt{q}} \quad \dots(v)$$

Where I_d is the DC current which is assumed to be a constant. The transformer rating on the valve side (in volt amperes) is given by

$$\begin{aligned} S_{tv} &= p \frac{E_m}{\sqrt{2}} I_v \\ &= p \frac{V_{do} \pi}{\sqrt{2} s q \sin \frac{\pi}{q}} \frac{I_d}{r\sqrt{q}} \\ &= \frac{\pi}{\sqrt{2}} \frac{V_{do} I_d}{\sqrt{q} \sin \frac{\pi}{q}} \quad \dots(vi) \end{aligned}$$

The transformer utilization factor $\frac{S_{tv}}{V_{do} I_d}$ is only a function of 'q'. The optimum value of q which results in maximum utilization is equal to 3. It is a fortunate coincidence at the AC power supply is 3 phase and the commutation group of 3 valves is easily arranged.

For $q = 3$,

$$\frac{S_{tv}}{V_{do} I_d} = 1.481 \quad \dots(vii)$$

The transformer utilization can be improved further if no valve groups can share a single transformer winding. In this case, the current (rms) rating of the winding can be increased by a factor of $\sqrt{2}$ while decreasing the number of windings by a factor of 2. For this case,

$$\frac{S_{tv}}{V_{do} I_d} = 1.047 \quad \dots(viii)$$

12 Explain the complete analysis of six pulse Graetz converter circuit with overlap for two valve conduction.

13. Due to the leakage inductance of the converter transformer and the impedance in the supply network, the current in a valve cannot change suddenly and thus commutation from one valve to the next cannot be instantaneous. For example, when valve 3 is fired, the current

transfer from valve 1 to valve 3 takes a finite period during which both valves are conducting. This is called overlap and its duration is measured by the overlap (commutation) angle 'u'. Each interval of the period of supply can be divided into two subintervals. In the first interval, three valves are conducting and in the second subinterval, two valves are conducting. This is based on the assumption that the overlap angle is less than 60° . As the overlap angle increases to 60° , there is no instant when only two valves are conducting. As the overlap angle increases beyond 60° , there is finite period during an interval, when four valves conduct and the rest of the interval during which three valves conduct. Thus there are three modes of the converter as follows:

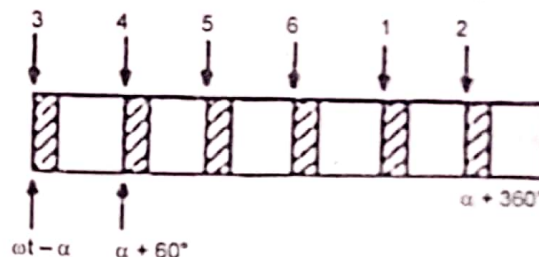


Fig. 1

Mode 1- Two and three valve conduction ($u < 60^\circ$)

Mode 2- Three valve conduction ($u = 60^\circ$)

Mode 3- Three and four valve conduction ($u > 60^\circ$)

(i) **Analysis of two and three valve conduction mode :**
For the interval considered, the bridge circuit can be reduced to that shown in fig. (2).

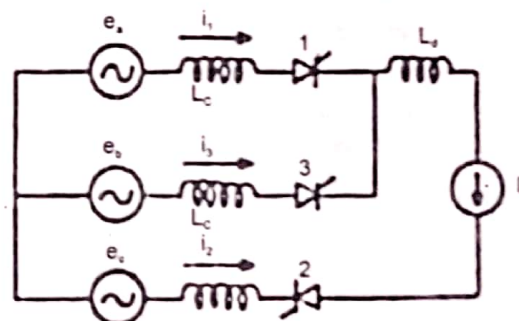


Fig. 2 : Equivalent circuit for 3 valve conduction

For this circuit,

$$e_b - e_a = L_c \left(\frac{di_3}{dt} - \frac{di_1}{dt} \right)$$

The L.H.S. in the above equation is called commutating emf whose value is given by

$$e_b - e_a = \sqrt{2} E_{LL} \sin \omega t$$

Which is also the voltage across valve 3 before it starts conducting. Since

$$i_1 = I_d - i_3$$

We get,

$$\sqrt{2} E_{LL} \sin \omega t = 2L_c \frac{di_3}{dt}$$

Solving the above equation, we get

$$i_3(t) = I_s (\cos \alpha - \cos \omega t), \alpha \leq \omega t \leq \alpha + u$$

Note that the solution is obtained from the initial condition

$$i_3(\omega t = \alpha) = 0$$

$$\text{At } \omega t = \alpha + u, i_3 = I_d$$

This gives

$$I_d = I_s [\cos \alpha - \cos(\alpha + u)]$$

$$I_s = \frac{\sqrt{2} E_{LL}}{2 \omega L_c}$$

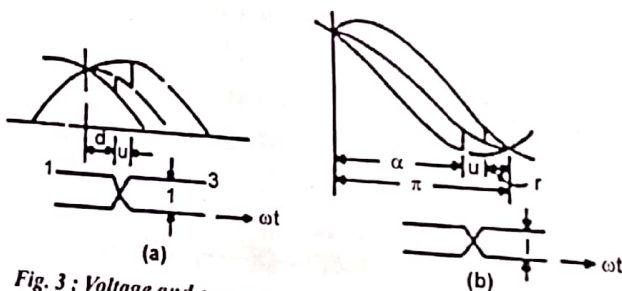


Fig. 3 : Voltage and current waveform during firing of a valve
(a) rectifier; (b) inverter

The waveforms of direct voltage and the valve currents during commutation for a rectifier and an inverter are shown in fig.(3(a)) and (3(b)) respectively. It is to be noted that during commutation, the instantaneous dc voltage is

$$\frac{-3}{2} e_c \text{ instead of } (e_b - e_c)$$

Average Direct Voltage : The average direct voltage can be obtained as

$$V_d = \frac{3}{\pi} \left[\int_{\alpha}^{\alpha+u} \frac{-3}{2} e_c d(\omega t) + \int_{\alpha+u}^{\alpha+60^\circ} (e_b - e_c) d(\omega t) \right]$$

$$= \frac{3}{\pi} \left[\int_{\alpha}^{\alpha+60^\circ} e_{bc} d(\omega t) - \int_{\alpha}^{\alpha+u} \frac{2e_b - e_c}{2} d(\omega t) \right]$$

$$= \frac{3}{\pi} \left[\int_{\alpha}^{\alpha+60^\circ} e_{bc} d(\omega t) - \int_{\alpha}^{\alpha+u} \frac{(e_b - e_a)}{2} d(\omega t) \right]$$

$$= V_{do} \cos \alpha - \frac{3}{2\pi} \int_{\alpha}^{\alpha+u} \sqrt{2} E_{LL} \sin \omega t d(\omega t)$$

$$= V_{do} \cos \alpha - \frac{3}{2\pi} \sqrt{2} E_{LL} [\cos \alpha - \cos(\alpha + u)]$$

Since $\frac{3}{\pi} \sqrt{2} E_{LL} = V_{do}$, we get

$$V_d = \frac{V_{do}}{2} [\cos \alpha + \cos(\alpha + u)]$$

$$V_d = V_{do} \left(\cos \alpha - \frac{I_d}{2I_s} \right) = V_{do} \cos \alpha - R_c I_d$$

$$R_c = \frac{3}{\pi} \omega L_c = \frac{3}{\pi} X_c$$

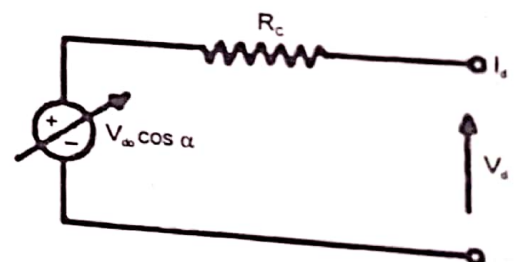


Fig. 4

R_c is called equivalent commutation resistance. It is something analogous to armature reaction in DC machines in the sense that it only represents a voltage drop and not a power loss. The equivalent circuit of the bridge rectifier is shown in the fig. (4).

Q.13 Explain the methods for obtaining the steady-state solution of equations for a six pulse HVDC converter bridge with filters.

Ans. Here the assumptions of constant DC current and constant, sinusoidal voltages at the AC bus are relaxed. The analysis takes into account impedances in both AC and DC networks. The steady-state solution of HVDC converter equation is periodic. With linear network elements, the equations are also piecewise linear. A method for obtaining

fast steady-state solution of the system equations for the general case is described.

Outline of the Method : The method is based on the following assumptions:

1. The system is described by sets of linear differential equations and each set is applicable for particular conduction pattern of the valves in bridge.
2. AC system is symmetrical and source voltages are balanced.
3. Firing pulses are generated at equal intervals of time.

Given the initial conditions and forcing functions, the solution can be obtained analytically cover each interval or sub-interval. The solution is periodic in steady-state, each period can be divided into p intervals, where p is the pulse number of the converter. Each interval, in general can be divided into two sub-intervals as follows:

1. $0 < t < t_1$, corresponding to the conduction of $(m+1)$ valves.
2. $t_1 < t < T_1$ corresponding to the conduction of m valves. ($T_1 = T/P$, T is the period of the AC voltage).

For example, in a 6-pulse converter, the normal mode consists of 3 and 2 valve conduction with $m=2$. The derivation of the steady-state solution proceeds from first computing t_1 from non-linear equation of the form

$$f(t_1) = 0$$

Once t_1 is obtained, the initial conditions can be calculated from sets of linear algebraic equations. These equations are derived from boundary conditions and symmetry considerations. In general, the following constraints apply in determining the boundary conditions.

Magnetic fluxes and electric charges must be continuous functions of time.

The current in the outgoing valve is zero at $t = t_1$

Development of the Method for Fast Steady-state Analysis : It is already mentioned, that there are two sub-intervals to be considered. Let the system be described by the following state equations.

$$\dot{x}_1(t) = [A_1]x_1(t) + [B_1]u_1(t), \quad 0 < t < t_1$$

$$\dot{x}_2(t) = [A_2]x_2(t) + [B_2]u_2(t), \quad t_1 < t < T_1$$

The orders of the state vectors x_1 and x_2 are $n+1$ and n respectively. Since the outgoing valve current becomes zero at t_1 , one state variable is eliminated in the second sub-interval. The solutions of the above equations are:

$$x_1(t) = x_{s1}(t) + e^{A_1 t} [x_1(0) - x_{s1}(0)]$$

$$x_2(t) = x_{s2}(t) + e^{A_2(t-t_1)} [x_2(t_1) - x_{s2}(t_1)]$$

Where $x_{si}(t)$ is the forced response in the sub-interval i . There is no loss of generality in assuming the current in the outgoing valve as x_1^{n+1} , superscript indicates the $(n+1)^{th}$ variable in the vector x_1 . Thus, we have

$$x_1^{n+1}(t_1) = 0 \Rightarrow Cx_1(t_1) = 0$$

Where $C = [0 \ 1]$ is a row vector. The vector $x_2(t)$ is the subset of the vector $x_1(t)$ with the last variable removed from $x_1(t)$. From the consideration of continuity in the state variables, we get $x_2(t_1) = [I_n \ 0] x_1(t_1) = [K] x_1(t_1)$

Where I_n is the identity matrix of n^{th} order. From considerations of symmetry, we have

$$x_1(0) = [S] x_2(T_1)$$

Where S is a constant matrix of dimension $(n+1)n$.

Q.14 With a neat diagram and waveforms explain the 6-pulse Graetz circuit without overlap.

Ans. Without Overlap : At any instant, two valves are conducting in the bridge, one from the upper commutation group and the second from the lower commutation group. The firing of the next valve in a particular group results in the turning off of the valve that is already conducting. This assumption that there is no overlap between the two valves in a group is incorrect. The valves are numbered in the sequence in which they are fired. Thus valve 2 is fired 60° after the firing of valve 1 and the valve 3 is fired 60° after the firing of the second valve. Each valve conducts for 120° and the interval between consecutive firing pulse is in 60° steady state.

The following assumptions are made to simplify the analysis :

- (i) The dc current is constant.
- (ii) The valves can be modelled as ideal switches with zero impedance when on (conducting) and with infinite impedance when off (not conducting).
- (iii) The AC voltages at the converter bus are sinusoidal and remain constant.

One period of the AC supply voltage can be divided into 6 intervals – each corresponding to the conduction of a pair of valves. The dc voltage waveform repeats for each interval. Thus, for the calculation of the average dc voltage, it is necessary to consider only one interval (say the interval

corresponding to the conduction of valves 2 and 3). Assuming the firing of valve 3 is delayed by an angle α , the instantaneous dc voltage V_d during the interval is given by

Average dc voltage

$$V_d = \frac{3}{\pi} \int_{\alpha}^{\alpha+60^\circ} \sqrt{2} E_{LL} \sin(\omega t + 60^\circ) d\omega t$$

$$= \frac{3}{\pi} \sqrt{2} E_{LL} [\cos(\alpha + 60^\circ) - \cos(\alpha + 120^\circ)]$$

$$V_d = \frac{3\sqrt{2}}{\pi} E_{LL} \cos \alpha = 1.35 E_{LL} \cos \alpha = V_{do} \cos \alpha$$

DC Voltage Waveform : The dc voltage waveform contains a ripple whose fundamental frequency is six times the supply frequency. This can be analysed in Fourier series and contains harmonics of the order.

$$h = np$$

Where p is the pulse number and n is an integer.

The rms value of the h^{th} order harmonic in dc voltage is given by

$$V_h = V_{do} \frac{\sqrt{2}}{h^2 - 1} [1 + (h^2 - 1) \sin^2 \alpha]^{1/2}$$

The waveforms of the direct voltage for different values of α are shown in the fig.(1). The waveforms of the valve voltage are also shown in the fig.(1).

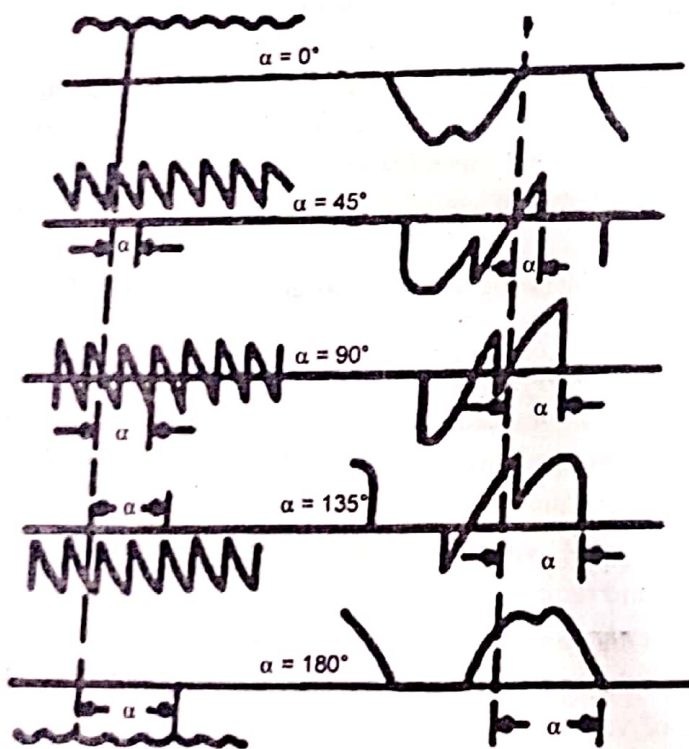


Fig. 1 : DC and valve voltage waveforms

AC Current Waveform : It is assumed that the direct current has no ripple (or harmonics). This is normally valid because of the smoothing reactor provided in series with the bridge circuit. The AC currents flowing through the valve (secondary) and primary windings of the converter transformer contain harmonics. The waveform of the current in a valve winding is shown in the fig.(2).

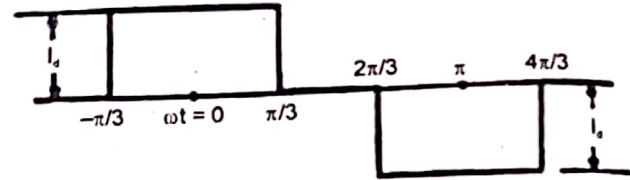


Fig. 2 : AC current waveform

The rms value of the fundamental component of current is given by

$$I_1 = \frac{1}{\sqrt{2}} \frac{2}{\pi} \int_{-\pi/3}^{\pi/3} I_d \cos \theta d\theta = \frac{\sqrt{6}}{\pi} I_d$$

Whereas the rms value of the current is

$$I = \sqrt{\frac{2}{3}} I_d$$

The harmonics contained in the current waveform of the order given by

$$h = np \pm 1$$

Where n is an integer, p is the pulse number. For pulse bridge converter, the order of AC harmonics are 7, 11, 13 and higher order. These are filtered out by using filters or each one of the first four harmonics and a high pass filter for the rest. The rms value of 11th harmonic is given by

$$I_h = I_1/h$$

The Power Factor : The AC power supplied to the converter is given by

$$P_{AC} = \sqrt{3} E_{LL} I_1 \cos \phi$$

Where $\cos \phi$ is the power factor. The DC power is equal to the AC power ignoring the losses in the converter. Thus we get,

$$P_{AC} = P_{DC} = V_{do} I_d = \sqrt{3} E_{LL} I_1 \cos \phi$$

Substituting for V_{do} and I_1 in the above equation we obtain

$$\cos \phi = \cos \alpha$$

The reactive power requirements are increased from zero (or reduced from 180°). When $\alpha = 90^\circ$, the power factor is zero and only reactive power is consumed.

Q.16 For a 12 pulse converter with $q = 4$, $s = 3$, $r = 1$, calculate the maximum DC power and transformer ratings (valve windings) if PIV rating of the valve is V and the rms current rating is I . Rework the problem if $q = 3$, $s = 4$, $r = 1$.

Ans. With even value of q , $PIV = 2E_m$

$$V_{do} = \frac{q}{\pi} s E_m \sin \frac{\pi}{q} = \frac{12}{\pi} \frac{V}{2} \sin \frac{\pi}{4} = \frac{3\sqrt{2}}{\pi} V = 1.35V$$

$$I = \frac{I_d}{r\sqrt{q}} = \frac{I_d}{2} \Rightarrow I_d = 2I$$

$$P_{d \max} = V_{do} I_d = 2.70VI$$

$$S_p = p \frac{E_m}{\sqrt{2}} I = \frac{12V}{2\sqrt{2}} I = 4.243VI$$

with $q = 3, s = 4, r = 1$

$$PIV = 2E_m \cos \frac{\pi}{2q} = 2E_m \cos 30^\circ = \sqrt{3}E_m$$

$$V_{do} = \frac{qs}{\pi} E_m \sin \frac{\pi}{q} = \frac{12}{\pi} \frac{V}{\sqrt{3}} \sin 60^\circ = \frac{6V}{\pi} = 1.91V$$

$$I_d = \sqrt{qr} I = \sqrt{3}I, P_{d \max} = V_{do} I_d = 1.91 \times \sqrt{3}VI = 3.31VI$$

$$S_{IV} = 12 \frac{E_m}{\sqrt{2}} I = \frac{12VI}{\sqrt{6}} = 4.9VI$$

with only six windings on the valve side

$$S_{IV} = \frac{4.9}{\sqrt{2}} VI = 3.465VI$$

Q.16 Explain the concept of PWM based HVDC system with waveforms.

Ans. Pulse Width Modulation : In order to control the power and reactive power, it is essential to control the magnitude and phase angle of the injected voltage by the converter. At the same time, it is desirable to regulate the DC bus voltage. Another requirement is to minimize the

injection of low order harmonics by the converter. The Pulse Width Modulation (PWM) is utilized to achieve the following objectives:

1. To control the output AC voltage for a constant DC voltage.
2. To minimize the harmonics subject to the constraints on the switching losses, generation of noise etc.

The frequency ratio

$$p = \frac{F}{f} \quad \dots(i)$$

Where F is the modulation frequency and f is the fundamental output frequency is an important parameter. p is also called as the pulse number. Higher the value of p , broader the spectrum of harmonics that can be eliminated. The switching frequency (F) is dependent on the devices used. The maximum switching frequency for GTO thyristors is only around 100 Hz. This is why PWM is not used in GTO based VSC employed in high power FACTS controllers. Multi-pulse or multilevel converters which require only one switching (on and off) in a cycle are used with GTO devices. On the other hand, IGBT devices can be operated up to 10 kHz. However, switching losses increase with frequency and are a limiting factor in the choice of F .

For $p > 20$, the modulating (carrier) wave and the reference (output) voltage need not be synchronized. It is not necessary that p be an integer. For comparing various modulation schemes, the modulation index (m) is normalized by taking the ratio of actual fundamental component of the modulated output voltage to that of the unmodulated output voltage, that is,

$$\bar{m} = \frac{E_{mod1}}{0.45V_{dc}} \quad \dots(ii)$$

Where E_{mod1} is the rms value of the fundamental component of the modulated output voltage.

Selective Harmonic Elimination (SHE) : This is also termed as the optimized PWM technique. By reversing the phase voltages a few times during each half cycle, it is possible to eliminate low order harmonics selectively. However, the high order harmonics may increase in magnitudes, but the current harmonics are not significantly affected due to the low pass filter characteristics of the AC system.

Figure shows the waveforms of the phase voltage (e_{aM}) with four and six reversals within one half cycle. The voltage reversals are affected at chosen instants such that the notches

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(caused by the voltage reversals) are placed symmetrically about the centre line of each half cycle. If there are 'p' switching (voltage reversals) in a quarter cycle, the rms value of the n^{th} harmonic voltage is given by,

$$E_n = \pm \frac{1}{\sqrt{2}} \frac{4}{\pi} \frac{V_{dc}}{2} \left[- \int_0^{\alpha_1} \sin n\theta d\theta + \int_{\alpha_1}^{\alpha_2} \sin n\theta d\theta \dots \pm \int_{\alpha_p}^{\pi/2} \sin n\theta d\theta \right]$$

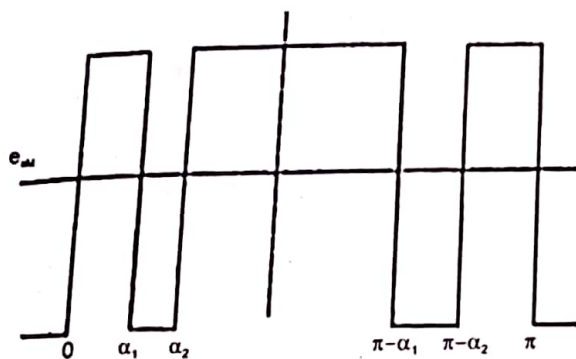
$$= \pm \frac{0.45}{n} V_{dc} [2(\cos n\alpha_1 - \cos n\alpha_2 + \cos n\alpha_3 - \dots) - 1] \dots \text{(iii)}$$

Where $\alpha_1, \alpha_2 \dots \alpha_p$ are switching angles within each quarter cycle. There are p degrees of freedom and are used to cancel (p - 1) harmonic components in the voltage and control the fundamental voltage. For example, if there are 3 switching at α_1, α_2 and α_3 , we can eliminate the fifth and seventh harmonics in addition to controlling the fundamental voltage. We get 3 transcendental equations given by

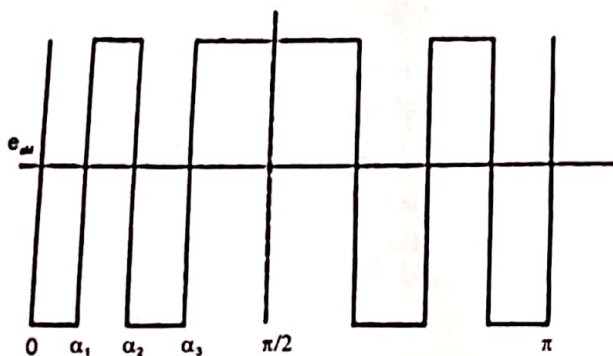
$$\bar{m} = 2[\cos \alpha_1 - \cos \alpha_2 + \cos \alpha_3] - 1 \dots \text{(iv)}$$

$$0 = 2[\cos 5\alpha_1 - \cos 5\alpha_2 + \cos 5\alpha_3] - 1 \dots \text{(v)}$$

$$0 = 2[\cos 7\alpha_1 - \cos 7\alpha_2 + \cos 7\alpha_3] - 1 \dots \text{(vi)}$$



(a) 4 Voltage reversals in a half cycle



(b) 6 voltage reversals in a half cycle

Fig. : Waveforms of e_M with voltage reversals

Note that although the phase voltage contain triplen harmonics, the line to line voltage are free from them.

PART-C

Q.17 Write down the analysis of Graetz bridge neglecting overlap in detail.

Ans. Analysis of Graetz Bridge Neglecting Overlap : In a LCC, the Graetz Bridge is connected to three balanced sinusoidal voltage sources on the AC side and constant DC current source on the DC side as shown in fig.(1).

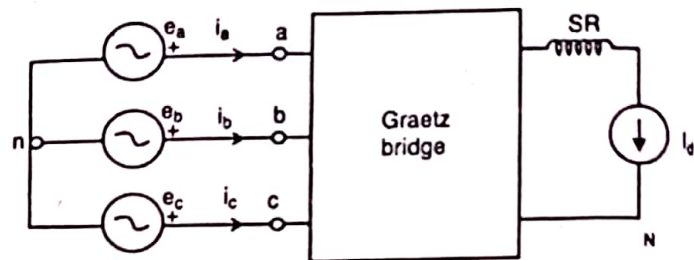


Fig. 1 : Line commutated converter

Each of the switches S_1 to S_6 in the Graetz bridge are made of series connected thyristor devices (to provide the required voltage rating). The valve is denoted by the thyristor symbol shown in fig.(2).

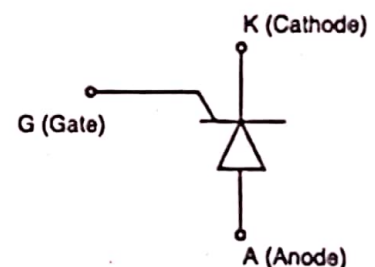


Fig. 2 : Thyristor symbol

A thyristor valve (ideally) can be viewed as a switch which can be turned on when the terminal A (anode) is positive with respect to the terminal K (cathode) and a gate (firing) signal is provided at the terminal G. Once the thyristor switch is turned on, it can be turned off only when the current through it goes to zero and there is a minimum commutation margin (when the voltage across the thyristor is reverse biased – the voltage at terminal A is negative with respect to K). In modern high voltage thyristor device (rated at 8 kV and above) the margin required may be hundreds of microseconds. In general, when no gate pulse is present, the switch should withstand both positive and negative voltages. Neglecting losses in the thyristor valves, the voltage across the device is

Note that the current can flow only in one direction through the thyristor switch (from anode to cathode) when it is turned on and the current through the devices is zero when it is not conducting.

Note that the current can flow only in one direction through the thyristor switch (from anode to cathode) when it is turned on. Thus, the switches S_1 , S_3 and S_5 carry current towards the terminal 'P' while the switches S_2 , S_4 , and S_6 carry current from the terminal 'N'. The direct current I_d has to flow through at least one valve in the upper group (S_1 , S_3 , and S_5) and one valve in the lower group (S_2 , S_4 and S_6). Neglecting overlap between any two valves in a group, we can state that only one valve conducts in the upper group or lower group. Thus in a cycle of the applied AC voltage, there are six equal intervals as shown in fig.(3).

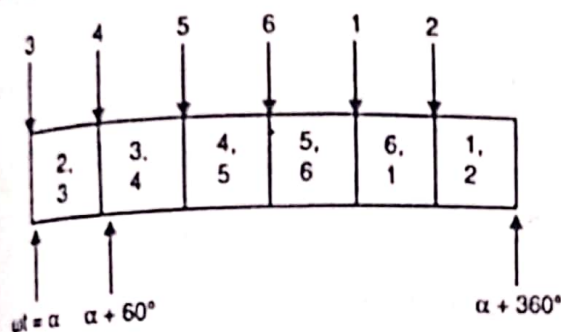


Fig. 3 : Timing diagram

We assume that the switch S_3 is turned on at $\omega t = \alpha$. Prior to this, switches S_1 and S_2 are conducting and we also assume that the switch S_1 turns off as soon as the switch S_3 turns on (neglecting the overlap). Overlap between the switches S_1 and S_3 cannot be neglected if there are inductors in the circuit (due to leakage reactance of the converter transformer). Table (1) shows the DC voltages across the converter (V_{PN}) for each of the six intervals.

Table 1 : DC bus voltage during each interval

Interval	I	II	III	IV	V	VI
Valves Conducting	2,3	3,4	4,5	5,6	6,1	1,2
Voltage (V_{PN})	e_{bc}	e_{ba}	e_{ca}	e_{cb}	e_{ab}	e_{ac}

Note that the voltage across the valve (switch S_3) is e_{ba} before it is fired (turned on). If we denote e_{ba}

$$e_{ba} = \sqrt{2}E_{LL} \sin \omega t \quad \dots(1)$$

The voltage across the switch S_3 is negative when $\omega t < 0$ and becomes positive when $\omega t > 0$. For a thyristor valve, we can delay the turn on by a delay angle (α) by applying a gate pulse at $\omega t = \alpha$. It is assumed that the delay angle is

constant in steady state.

The expression for e_{ba} given in equation (1) implies that

$$\left. \begin{aligned} e_a(t) &= \sqrt{\frac{2}{3}}E_{LL} \sin(\omega t + 150^\circ) \\ e_b(t) &= \sqrt{\frac{2}{3}}E_{LL} \sin(\omega t + 30^\circ) \\ e_c(t) &= \sqrt{\frac{2}{3}}E_{LL} \sin(\omega t - 90^\circ) \end{aligned} \right\} \quad \dots(2)$$

Note that we assume only positive sequence supply voltages. The expressions for the DC voltage (V_{PN}) and voltage across valve 1 for each interval, are shown in table 2. The waveforms of these voltages for $\alpha = 0^\circ, 45^\circ, 90^\circ, 135^\circ$ and 180° are shown in fig.(4). The DC voltage waveform shows that there are six pulsations (cycles of ripple) per cycle of the AC voltage. The pulse number (which is defined as the ratio of the base frequency of the DC voltage ripple to the fundamental frequency of the AC voltage) for Graetz bridge is six. We also note that the average DC voltage reduces (and ripple content increases) as α is increased from 0° to 90° . On the other hand, as α is increased from 90° to 180° , the average DC voltage (in the negative direction) increases and the ripple content reduces.

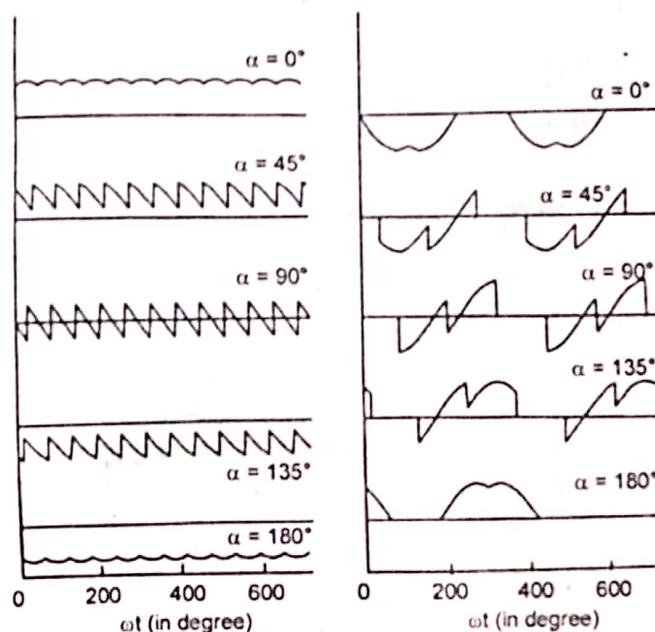


Fig. 4 : DC and valve voltage waveforms

Table 2 : DC and Valve Voltages

Interval No.	Conducting Valves	DC Voltage	Voltage Across Valve 1
I	2, 3	$\sqrt{2}E_{LL} \sin(\omega t + 60^\circ)$	$e_{ab} = -\sqrt{2}E_{LL} \sin(\omega t)$
II	3, 4	$\sqrt{2}E_{LL} \sin(\omega t)$	$e_{ab} = -\sqrt{2}E_{LL} \sin(\omega t)$
III	4, 5	$\sqrt{2}E_{LL} \sin(\omega t - 60^\circ)$	$e_{ac} = \sqrt{2}E_{LL} \sin(\omega t + 120^\circ)$
IV	5, 6	$\sqrt{2}E_{LL} \sin(\omega t - 120^\circ)$	$e_{ac} = \sqrt{2}E_{LL} \sin(\omega t + 120^\circ)$
V	6, 1	$\sqrt{2}E_{LL} \sin(\omega t - 180^\circ)$	0
VI	1, 2	$\sqrt{2}E_{LL} \sin(\omega t - 240^\circ)$	0

The DC voltage waveform repeats for each interval and hence average DC voltage can be found by taking the average of V_{PN} over one interval during which, a pair of valves conduct. This is based on the assumption that the applied voltages are balanced, sinusoidal (defined by equation (2) an dE_{LL} remains constant (over a cycle of the supply frequency). We also assume that the firing of the valves is equidistant (at the interval of 60°). This also implies that each valve conducts for 120° duration.

Let us consider the interval corresponding to the conduction of valves 2 and 3. At the beginning of the interval valve 3 is fired and at the end of the interval, valve 2 turns off.

Assuming the firing of valve 3 is delayed by an angle α , (that is α° after the zero crossing of the commutation voltage for valve 3-voltage e_{ba}), the instantaneous DC voltage V_d during the interval is given by

$$V_d = V_{PN} = e_b - e_c = e_{bc}, \alpha \leq \omega t \leq \alpha + 60^\circ$$

$$\text{If, } e_{ba} = \sqrt{2}E_{LL} \sin \omega t \text{ from equation (1)}$$

$$\text{then, } e_{bc} = \sqrt{2}E_{LL} \sin(\omega t + 60^\circ)$$

$$\text{Average dc voltage} = V_d = \frac{3}{\pi} \int_{\alpha}^{\alpha+60^\circ} \sqrt{2}E_{LL} \sin(\omega t + 60^\circ) d\omega t$$

$$= \frac{3}{\pi} \sqrt{2}E_{LL} [\cos(\alpha + 60^\circ) - \cos(\alpha + 120^\circ)]$$

$$V_d = \frac{3}{\pi} \sqrt{2}E_{LL} \cos \alpha = 1.35 E_{LL} \cos \alpha$$

$$= V_{do} \cos \alpha$$

...(3)

Equation (3) indicates that for different values of V_d is variable. The range of α is 180° (from 0° to 180°), correspondingly V_d can vary from V_{do} to $-V_{do}$. Thus same converter can act as rectifier or inverter depend upon whether the dc voltage is positive or negative.

It is to be noted that this is based on the assumption of continuous conduction of current by any valve over the 1 interval. Note that the DC current flow is unidirectional.

DC Voltage Harmonics :

The dc voltage waveform contains a ripple whose fundamental frequency is six times the supply frequency. It can be analysed in Fourier series and contains harmonics of the order

$$h = 6n$$

Where n is an integer.

The harmonic (rms value) in the dc voltage is given by

$$V_h = \frac{\sqrt{a_h^2 + b_h^2}}{\sqrt{2}}$$

$$\text{Where, } a_h = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} V_d \cos(h\theta) d\theta,$$

$$b_h = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} V_d \sin(h\theta) d\theta$$

It can be shown that the rms value of the h^{th} harmonic in dc voltage is given by,

$$V_h = V_{do} \frac{\sqrt{2}}{h^2 - 1} [1 + (h^2 - 1) \sin^2 \alpha]^{1/2}$$

Valve Voltage Waveform : Figure (4) also shows the valve voltage waveform which shows three voltage jumps that arise due to the commutation from one valve to the next in the same commutation group. The valve voltage waveform can be obtained from the expressions for a valve voltage given in table (2) (Note that the waveform is similar for all valves). The voltage jumps are all of the same magnitude given by

$$V_j = \sqrt{2} E_{LL} \sin \alpha = (\pi/3) V_{do} \sin \alpha \quad \dots(5)$$

These voltage jumps contribute to the voltage stresses (dv/dt) across the valve. Valve damping circuits are provided to relieve these stresses.

Although α can vary from 0° to 180° , the full range cannot be utilized. In order to ensure the firing of all the series connected thyristors, it is necessary to provide a minimum limit of α greater than zero, (say 5°). Also, in order to allow for the turn-off time of a valve, it is necessary to provide an upper limit less than 180° . The delay angle α is not allowed to go beyond $(180^\circ - \gamma)$ where γ is called the extinction angle (sometimes also called the margin angle). The minimum value of the extinction angle is typically 10° , although in normal operation as an inverter, it is not allowed to go below 15° or 18° .

In the absence of forced commutation, α cannot be made negative.

AC Current Waveform : It is assumed that the direct current has no ripple (or harmonics). This is normally valid because of the smoothing reactor provided in series with the bridge circuit. The AC currents flowing through the valve (secondary) and primary windings of the converter transformer contain harmonics. The waveform of the current in a valve winding (say in phase b) is shown in fig. (5).

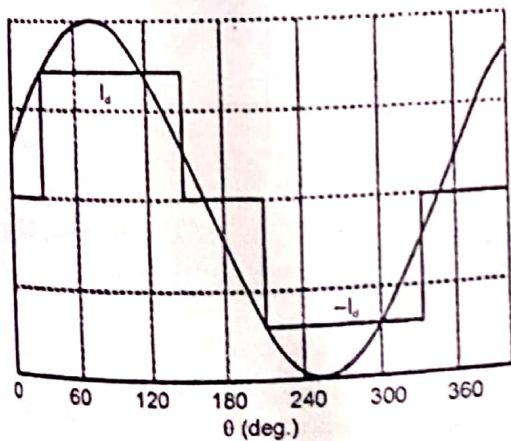


Fig. 5 : Waveforms of i_b and e_b

The current in a valve winding is determined by the currents flowing in the valves connected to that winding. For

example, the current in the phase 'b', i_b (see fig. (1)) is i_3 when valve 3 is conducting and $-i_6$ when valve 6 is conducting. Note that each valve conducts for 120° and valve 6 is turned on 180° after the firing of valve 3. Figure (5) also shows the applied voltage, $e_b = \sqrt{2} E_{LL} \sin(\omega t + 30^\circ)$.

From equations (1) and (2), we have

$$i_b = i_3 = I_d, \alpha \leq \omega t \leq \alpha + 120^\circ$$

$$= -i_6 = -I_d, \alpha + 180^\circ \leq \omega t \leq \alpha + 300^\circ$$

Defining $\theta = \omega t - \alpha - 60^\circ$, the rms value of the fundamental component is given by

$$I_1 = \frac{1}{\sqrt{2}} \frac{2}{\pi} \int_{-\pi/3}^{\pi/3} I_d \cos \theta d\theta = \frac{\sqrt{6}}{\pi} I_d \quad \dots(6)$$

Whereas rms value of the current is,

$$I = \sqrt{\frac{2}{3}} I_d \quad \dots(7)$$

The harmonics contained in the waveform are of the order given by

$$h = 6n \pm 1$$

Where n is an integer. Thus, the order of AC harmonics are 5, 7, 11, 13 and higher. These are filtered out by using tuned filters for each one of the first four harmonics and a high pass filter for the rest. The rms value of the h^{th} harmonic is given by

$$I_h = \frac{I_1}{h} \quad \dots(8)$$

The Power Factor : The AC power supplied to the converter is given by

$$P_{ac} = \sqrt{3} E_{LL} I_1 \cos \phi$$

Where ϕ is the (fundamental) frequency power factor.

The DC power must match the AC power ignoring the losses in the converter. Thus, we get

$$P_{ac} = P_{dc} = V_d I_d = \sqrt{3} E_{LL} I_1 \cos \phi \quad \dots(9)$$

Substituting for V_d and I_1 from equations (3) and (6) in the above equation, we obtain

$$\cos \phi = \cos \alpha \quad \dots(10)$$

The reactive power requirements are increased as α is increased from 0 (or reduced from 180°). When $\alpha = 90^\circ$,

the power factor is zero and only reactive power is consumed.

Q.18 Explain the complete characteristics of 12-pulse converter.

Ans. 12-Pulse Converter : In HVDC transmission, the series conduction of converter groups has been preferred because of the ease of control and protection as well as the requirements of high voltage rating. Thus a 12-pulse converter is obtained by the series connection of two bridges. The 30° phase displacement between the two sets of source voltages is achieved by the transformer connections Y/Y for feeding one bridge and Y/ Δ for feeding the second bridge.

The use of 12-pulse converter is preferable over the six pulse converter because of the reduced filtering requirements. However, increase in pulse number beyond 12 is not practical because the non-characteristic harmonics (which arise due to asymmetry in firing, imbalance in supply etc.) are not eliminated.

Analysis of a 12-Pulse Converters : To analyze the operation of a 12-pulse converter, consider the converter circuit diagram shown in fig.(1). This shows two transformers with primary windings connected in star and connected to the common supplied voltage. The currents supply by the supply are the sums of the currents flowing in the two primary windings. Thus,

$$\left. \begin{aligned} i_A &= i_{AS} + i_{AD} \\ i_B &= i_{BS} + i_{BD} \\ i_C &= i_{CS} + i_{CD} \end{aligned} \right\} \dots(1)$$

In determining the voltage and current waveforms there is no loss of generality, in assuming that the turns ratio of the transformer feeding bridge 1 as 1:1 and the turns ratio of the transformer feeding the second bridge as $1:\sqrt{3}$. Assuming ideal transformers, we will get the identical magnitude of the AC voltages applied to the two bridges. Assuming e_A , e_B and e_C are defined as follows:

$$e_A = e_{AS} = \sqrt{\frac{2}{3}} E_{LL} \sin(\omega t + 150^\circ)$$

$$e_B = e_{BS} = \sqrt{\frac{2}{3}} E_{LL} \sin(\omega t + 30^\circ)$$

$$e_C = e_{CS} = \sqrt{\frac{2}{3}} E_{LL} \sin(\omega t - 90^\circ)$$

We get the line to line voltages applied to the second bridge as

$$e_{baD} = -\sqrt{3}e_A = \sqrt{2}E_{LL} \sin(\omega t - 30^\circ) \dots$$

We note that the voltages applied to bridge 2 lag the voltages applied to the bridge 1 by 30° . Assuming the identical delay angle for both the bridges, valve 3 in the second bridge will be fired 30° after firing the valve 3 in bridge 1.

Thus, in a 12 pulse converter, there are 12 intervals a cycle, each of duration 30° . In each interval, 4 valves (each from the two bridges) conduct. Table shows the conducting valves from each bridge for all the intervals (note that the subscripts S and D denote the bridges with star connected and delta connected windings respectively).

Table : Bridge voltages in each interval

Interval	I(a)	I(b)	II(a)	II (b)	III (a)	III (b)	IV(a)	IV(b)	V(a)	V(b)	VI(a)	VI(b)
Conducting Valves Bridge 1	2, 3		3, 4		4, 5		5, 6		6, 1		1, 2	
V_{d1}	e_{bcS}		e_{baS}		e_{caS}		e_{cbS}		e_{abS}		e_{acS}	
Conducting Valves Bridge 2	1, 2	2, 3	3, 4		4, 5		5, 6		6, 1		1, 2	
V_{d2}	e_{acD}	e_{bcD}	e_{baD}		e_{caD}		e_{cbD}		e_{abD}		e_{acD}	

The DC voltage V_{dc} for a 12 pulse converter is the sum of the DC voltages for the individual bridges. Thus,

$$V_d = V_{d1} + V_{d2} \dots(3)$$

For the interval I(a), the R.H.S. can be expressed as

$$V_d = e_{bcS} + e_{acD}$$

$$= \sqrt{2}E_{LL} \sin(\omega t + 60^\circ) - \sqrt{2}E_{LL} \sin(\omega t - 270^\circ)$$

$$= \sqrt{2}E_{LL} [\sin(\omega t + 60^\circ) + \cos \omega t]$$

$$= E_m \sin(\omega t + 75^\circ) \quad \dots(4)$$

Where E_m is defined as

$$E_m = \sqrt{2} E_{LL} (2 \cos 15^\circ) = \sqrt{2} E_{LL} (1.93185) \quad \dots(5)$$

The average DC voltage (V_{dc}) is given by

$$V_{dc} = \frac{6}{\pi} \left[\int_{\alpha}^{\alpha+30^\circ} E_m \sin(\omega t + 75^\circ) d\omega t \right] \quad \dots(6)$$

$$= 2 V_{do} \cos \alpha$$

Where,

$$V_{do} = \frac{3}{\pi} \sqrt{2} E_{LL}$$

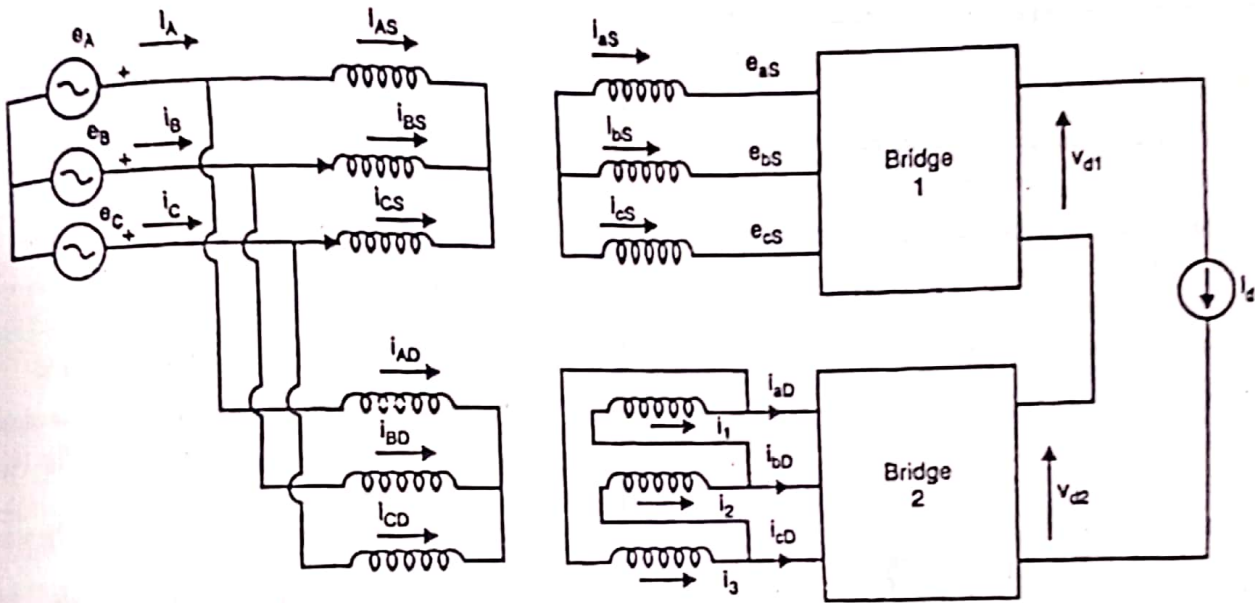


Fig. 1 : A 12 pulse converter

The current i_{BD} flowing in the primary winding of the star/delta connected transformer (feeding bridge 2 in Fig.1) is given by

$$i_{BD} = \sqrt{3} i_2 \quad \dots(7)$$

Where i_2 is the current flowing in the secondary winding that is coupled with phase B. Assuming that $i_1 + i_2 + i_3 = 0$, we can show that

$$i_{BD} = \frac{1}{\sqrt{3}} (i_{bD} - i_{cD}) \quad \dots(8)$$

The waveform of i_{BS} , i_{BD} and i_B are shown in fig.2. In deriving the waveform of i_{BD} , it is to be noted that i_{bD} lags i_{bS} by 30° and i_{cD} lags i_{bD} by 120° .

It can be shown that the rms value of the fundamental component of the supply current in 12 pulse converter is given by

$$I_1 = 2 \cdot \frac{\sqrt{6}}{\pi} I_d \quad \dots(9)$$

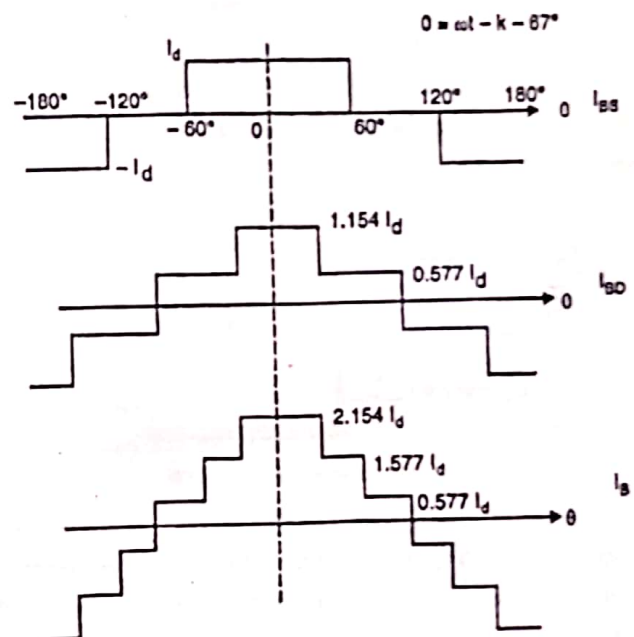


Fig. 2 : Waveforms of current in 12 pulse converter

The harmonic current I_h is given by,

$$I_h = \frac{I_1}{h}$$

Where, $h = 12n \pm 1$, n is an integer

$= 6m \pm 1$, m is an even integer.

In comparing with the harmonic currents of a six pulse converter, it is obvious that the fifth and seventh harmonics flowing in the two transformers cancel each other. This can be generalized to the case whenever m is an odd integer.

Q.19 With neat sketch explain the concept of VSC based HVDC system.

Ans. Voltage Source Converter :

1. Basic Two Level (Graetz Bridge) Converter : The configuration of the Voltage Source Converter (VSC) with two levels is as shown in fig.(2) with six switches. However, each switch consists of a self commutated IGBT device and an antiparallel connected diode as shown in fig.(1).

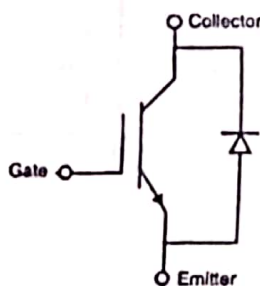


Fig. 1 : Switch consisting of IGBT and diode

The DC side is connected to the capacitor and the AC side is connected to the source voltage through inductors as shown in fig.(2).

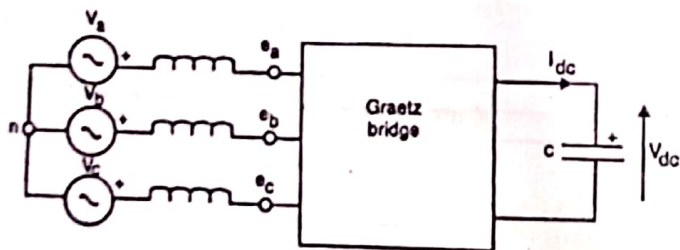


Fig. 2 : A voltage source converter

Here, the voltage e_a , e_b and e_c are injected by the VSC and depend on V_{dc} , the voltage across the capacitor. The current i_{dc} flowing into the capacitor is a function of the source currents i_a , i_b and i_c .

In each leg of the converter, only one switch can be turned on at any given time to avoid short circuit across the

capacitor. Each switch conducts for 180° and switch 1 must be turned off before switch 4 is turned on in the leg connected to phase 'a'. Similar comments apply for switches in the legs connected to phases 'b' and 'c'. The waveform of the voltage e_a depends on the reference point. Figure (3) shows the leg connected to phase 'a'.

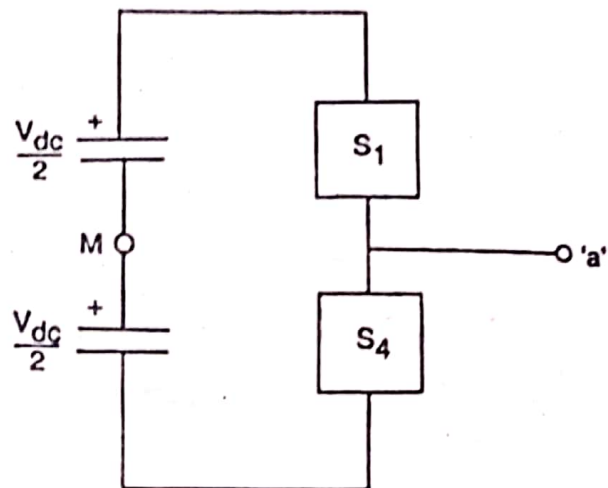
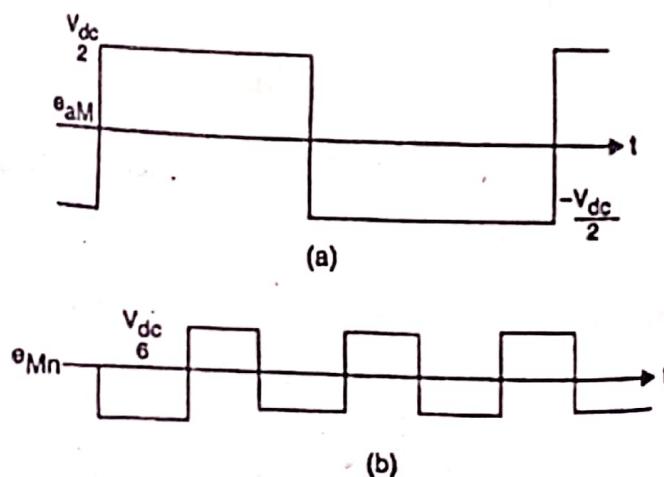


Fig. 3 : A VSC leg connected to phase 'a'

With respect to the midpoint 'M' of the capacitor, the waveform of the voltage e_{aM} is shown in fig. (4(a)). This voltage has only 2 levels $+\frac{V_{dc}}{2}$ and $-\frac{V_{dc}}{2}$. The voltages e_b and e_c lag e_{aM} by 120° and 240° respectively.

It can be shown that e_{Mn} (voltage of midpoint M with respect to the supply neutral) consists of only a zero sequence voltage with triplen harmonics (see Fig. (4(b))). Subtracting e_{Mn} from e_{aM} , we get e_{an} (the injected voltage with respect to supply neutral). The waveform of e_{an} is shown in fig. (4(c)). The phase to phase voltage e_{ab} does not contain zero sequence component and its waveform is shown in fig. (4(d)). Both e_{an} and e_{ab} contain only harmonics of the order $h = 6k \pm 1$, k is an integer.



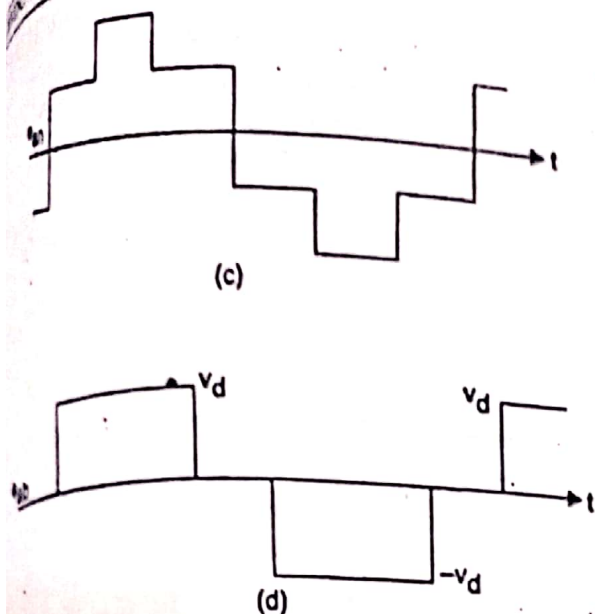


Fig. 4 : Waveforms of e_{aM} , e_{MN} , e_{aN} and e_{ab}

The peak value of the fundamental components of the injected voltage e_a is given by

$$E_1 = \frac{4}{\pi} \left(\frac{V_{dc}}{2} \right) = \sqrt{2} (0.90) \left(\frac{V_{dc}}{2} \right) \quad \dots(1)$$

0.90 is the rms value of the fundamental component of square wave with an amplitude of 1.0. The harmonic voltage injected is given by

$$E_h = \frac{E_1}{h} \quad \dots(2)$$

A Three Level Voltage Source Converter : A three level VSC (also termed as Neutral Point Clamped (NPC) converter) is a special case of multilevel diode-clamped converter. This has two capacitors on the DC bus and four valves (switches) in each phase leg as shown in fig.(5).

There are two clamping diodes in each phase. Although the number of switches is doubled, the voltage rating is also doubled. For the same level of the capacitor voltage, the number of devices required per switch are halved (as compared to a two level converter).

The operation of a three level converter is determined by the switching rule that only two valves that are directly connected in series can be switched on at a time. In fig.(5), switching on S_1 and S_2 connects the phase to the positive terminal P, while S_2 and S_3 connects to the midpoint M via clamping diodes, S_3 and S_4 connects to the terminal N. Thus, the resulting voltage injected (e_{aM}) has three levels (including zero level) as shown in fig.(6).

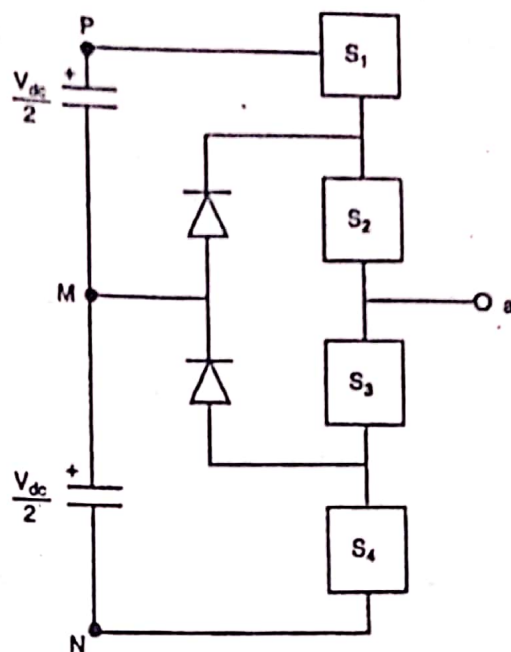


Fig. 5 : A three level VSC leg connected to phase 'a'

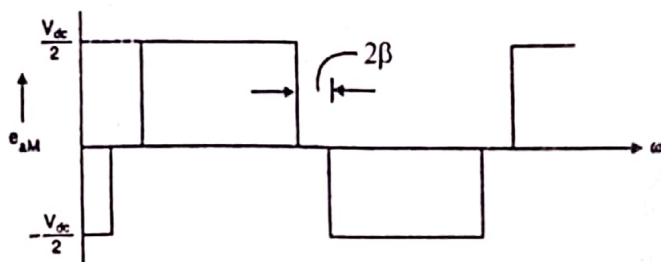


Fig. 6 : Waveform of e_{aM} in a three level converter

It can be shown that the peak value of the fundamental components of the injected voltage e_a is given by

$$E_1 = \frac{4}{\pi} \left(\frac{V_{dc}}{2} \right) \cos \beta = \frac{2}{\pi} V_{dc} \cos \beta \quad \dots(3)$$

The harmonic voltage (peak value) is given by

$$E_h = \frac{2}{h\pi} V_{dc} \cos(h\beta), h = 6k \pm 1 \quad \dots(4)$$

Where k is an integer.

For $\beta = 15^\circ$, three level converter behaves like a quasi 12 pulse converter with low values of the 5th and 7th harmonics. The rms value of the fundamental component is $0.435 V_{dc}$. (For a two level converter the rms value of the fundamental components is $0.45 V_{dc}$).

In comparison with a two level converter, a three level converter provides better performance. However, it results in more complex layout resulting in larger footprint and higher costs.

Q.20 Explain the effect of finite smoothing reactor in HVDC transmission system.

Ans. Effect of Finite Smoothing Reactor : Assumed that I_d is constant which implies that the reactance of the smoothing reactor is very high (theoretically infinite). If the smoothing reactor is finite, then we cannot neglect the ripple in the DC current. The average DC current (I_d) should be above a minimum value, (corresponding to the peak value of the ripple) such that the current in the DC link remains continuous. Since valve currents can be only unidirectional, when the current goes to zero, there is discontinuity. The extinction of the link current can lead to over voltages across the valves due to the lightly damped L-C circuit consisting of smoothing reactor and the line capacitance.

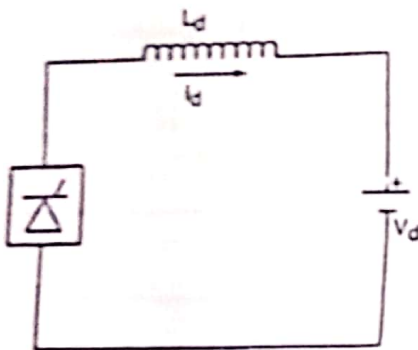


Fig. 1 : Equivalent circuit for calculating ripple

To compute the ripple in the DC link current, consider the equivalent circuit in fig.1. Here L_d is the inductance of the reactor and V_d is the constant voltage source that represents the effect of the converter at the other terminal. The time varying voltage $v_d(t)$ across the converter is given by

$$v_d(t) = E_m \cos(\omega t), \quad \frac{-\pi}{p} + \alpha \leq \omega t \leq \frac{\pi}{p} + \alpha \quad \dots(1)$$

Since, the losses in the circuit are neglected, we also have the constraint given by

$$V_d = \frac{p}{2\pi} \int_{\frac{-\pi}{p} + \alpha}^{\frac{\pi}{p} + \alpha} v_d(t) d\omega t \quad \dots(2)$$

Substituting equation (1) in (2), we get

$$V_d = \frac{p}{\pi} E_m \sin \frac{\pi}{p} \cos \alpha \quad \dots(3)$$

$v_d(t)$ and V_d are shown in fig.(2). It is obvious that current is minimum when $v_d(t) = V_d$ and is increasing. For $\alpha = 0$, the instant of minimum current occurs after the firing instant. However, it can be observed that as α increases, the current minimum occurs at the firing instant. The minimum value ($\bar{\alpha}$) for which the minimum current occurs at the firing instant, is given by the equation.

$$V_d = E_m \cos \left(\bar{\alpha} - \frac{\pi}{p} \right) \quad \dots(4)$$

Substituting equation (3) in equation (4), we can show that

$$\tan \bar{\alpha} = \frac{\pi}{p} - \cot \frac{\pi}{p} \quad \dots(5)$$

For $p = 6$, $\bar{\alpha} \approx 10^\circ$ and for $p = 12$, $\bar{\alpha} \approx 5^\circ$. Since, α is normally greater than $\bar{\alpha}$ in steady state, we will compute the ripple current for this case.

The expression for the direct current, i_d in fig.(1) obtained as

$$i_d = \frac{1}{\omega L_d} \int_{\alpha - \frac{\pi}{p}}^{\omega t} (v_d - V_d) d\omega t + i_{d \min} \quad \dots(6)$$

Substituting for v_d from equation 1, we get

$$i_d = \frac{1}{\omega L_d} \left[E_m \sin \omega t - E_m \sin \left(\alpha - \frac{\pi}{p} \right) - V_d \left(\omega t + \frac{\pi}{p} - \alpha \right) \right] + i_{d \min} \quad \dots(7)$$

The average current I_d in the DC link is given by

$$I_d = \frac{p}{2\pi} \int_{\frac{-\pi}{p} + \alpha}^{\frac{\pi}{p} + \alpha} i_d d\omega t \quad \dots(8)$$

Substituting for i_d from equation (7) in equation (8) we can derive

$$I_d = \left(\frac{p}{\pi} E_m \sin \frac{\pi}{p} \right) \frac{1}{\omega L_d} \left(1 - \frac{p}{\pi} \cot \frac{\pi}{p} \right) \sin \alpha + i_{d \min} \quad \dots(9)$$

$$= \frac{p}{6} \frac{V_{do}}{\omega L_d} \left(1 - \frac{\pi}{p} \cot \frac{\pi}{p} \right) \sin \alpha + i_{d \min}$$

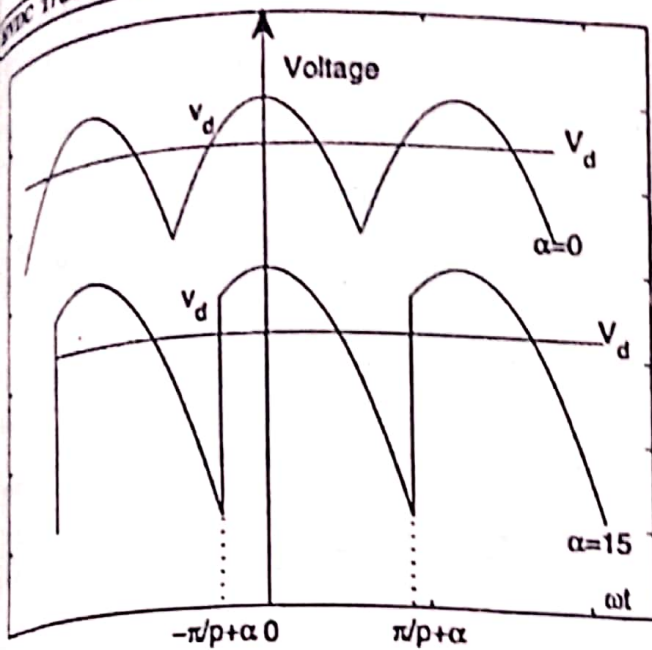


Fig. 2 : Waveforms of $v_d(\omega t)$ and V_d

Where,

$$V_{do} = \frac{3\sqrt{2}}{\pi} E_{LL} = \frac{6}{p} \left(\frac{p}{\pi} E_m \sin \frac{\pi}{p} \right)$$

For the current in the DC link to be continuous, $i_{dmin} \geq 0$.
For the limiting case of the $i_{dmin} = 0$, we get

$$I_{dmin} = \frac{p}{6} \frac{V_{do}}{\omega L_d} \left(1 - \frac{\pi}{p} \cot \frac{\pi}{p} \right) \sin \alpha$$

The factor $\left(1 - \frac{\pi}{p} \cot \frac{\pi}{p} \right)$ has the value of 0.0931 for $p = 6$ and 0.0230 for $p = 12$.

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